Power Supply Systems

Part II

Phase 9: Substation and Busbar Arrangement

Aim: To learn substations, substation equipment, pole-mounted substation and busbar schemes from absolute zero.

Style: Simple English, SN ma'am style, clean diagrams, no overlapping labels, every diagram word explained.

Target: Definitions, classifications, equipment functions, single-line diagrams, pole-mounted substation, busbar arrangements and semester-ready answers.

Subject:	Power Supply Systems
Section:	Part II
Phase:	9 – Substation and Busbar Arrangement
Level:	Absolute zero to semester-question-ready

Academic article style notes

Contents

1 Phase 9 Roadmap

Phase 9 Goal

After protection devices, we now study the place where switching, transformation, protection and distribution are practically arranged: the **substation**.

Phase 9 Learning Flow

Substation Meaning and Functions
↓
Classification of Substations
↓
Major Substation Equipment
↓
Single-Line Diagram
↓
Pole-Mounted Substation
↓
Busbar Arrangements

How to Read the Diagram

Word	**Meaning**
Substation	A place where voltage is transformed, circuits are switched, protected and controlled.
Classification	Types of substations according to function, voltage and construction.
Major Equipment	Transformer, busbar, circuit breaker, isolator, CT, PT, lightning arrester and earthing.
Single-Line Diagram	Simplified diagram showing main electrical connections using one line.
Pole-Mounted Substation	Small outdoor distribution substation mounted on poles.
Busbar Arrangements	Different ways of connecting incoming and outgoing circuits to busbars.

2 Meaning of Substation

Definition

A **substation** is a part of the power system where voltage is transformed from one level to another, and where switching, protection, control and distribution of electrical power are carried out.

A substation acts like a control centre between:

$$\boxed{\text{Transmission system} \rightarrow \text{Substation} \rightarrow \text{Distribution system}}$$

Memory Hook

Substation does four main jobs:

Transform voltage | Switch circuits | Protect equipment | Distribute power

3 Functions of a Substation

The main functions of a substation are:

1. To step up or step down voltage using transformers.
2. To switch transmission and distribution circuits.
3. To protect lines and equipment during faults.
4. To isolate faulty sections.
5. To measure voltage, current, energy and power.
6. To improve reliability of supply.
7. To distribute power to outgoing feeders.
8. To provide earthing and lightning protection.

Semester Answer Style

For a 4-mark answer, write definition of substation and any six functions.

4 Classification of Substations

Substations may be classified in different ways.

4.1 According to Service Requirement

Type	**Meaning**
Transformer substation	Voltage level is changed using transformers.
Switching substation	Switching operation is performed without changing voltage level.
Converting substation	AC is converted to DC or DC to AC.
Frequency changing substation	Frequency is changed from one value to another.
Power factor correction substation	Power factor is improved using capacitor banks or synchronous condensers.

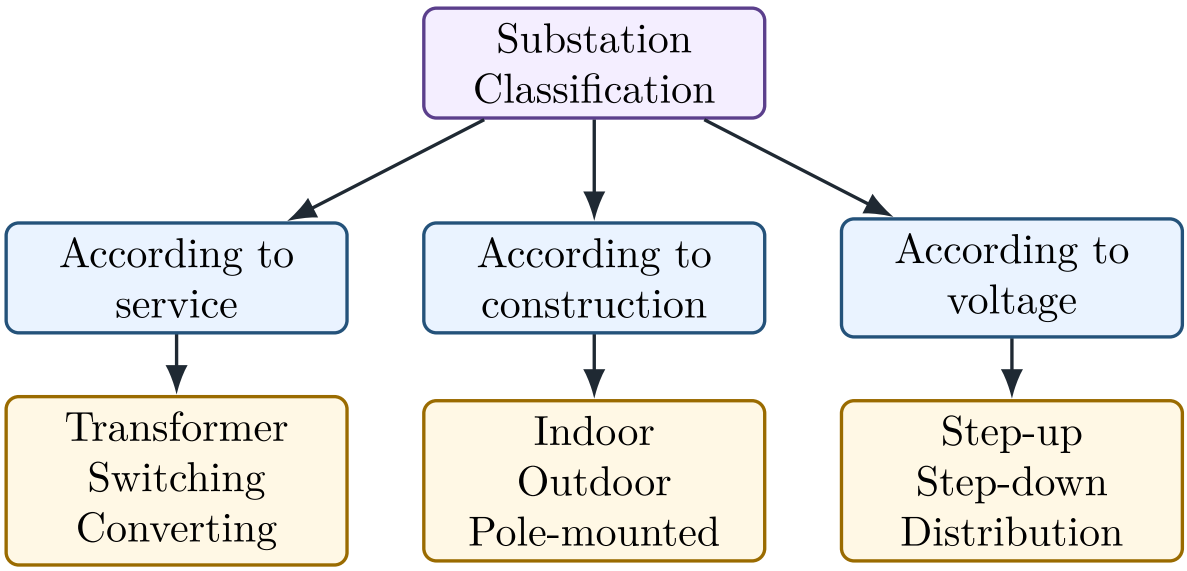
4.2 According to Construction

Type	**Meaning**
Indoor substation	Equipment is installed inside a building. Used for lower voltages or space-limited areas.
Outdoor substation	Equipment is installed in open air. Used for high-voltage substations.
Pole-mounted substation	Distribution transformer and accessories are mounted on poles.
Underground substation	Installed below ground level where land is limited.
Gas-insulated substation	Uses SF_6 gas insulation and needs less space than air-insulated substation.

4.3 According to Voltage Transformation

Type	**Meaning**
Step-up substation	Increases voltage for transmission. Usually located near generating station.
Step-down substation	Decreases voltage for sub-transmission or distribution.
Distribution substation	Steps down voltage to distribution level, such as 11 kV to 400/230 V.

5 Classification Flowchart



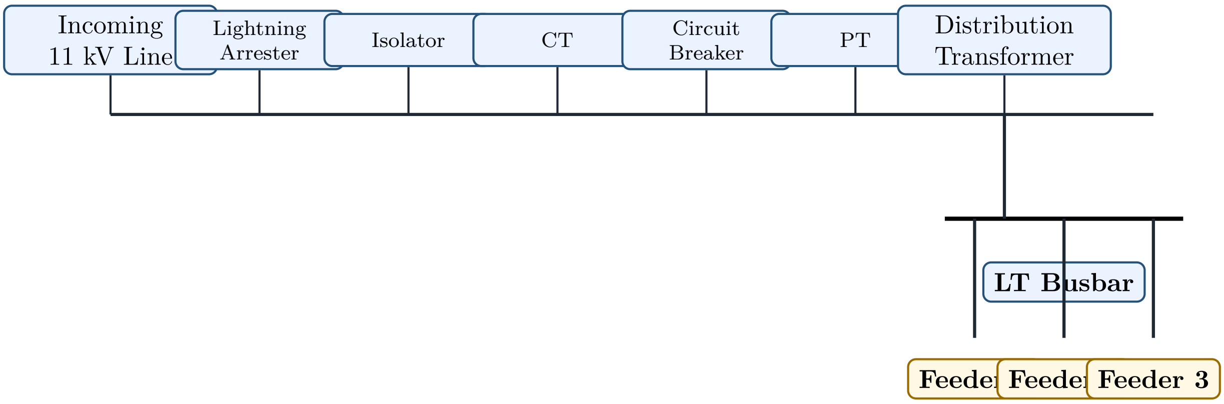
How to Read the Diagram

Word	Meaning
Substation Classification	Different ways of grouping substations.
According to service	Classification based on what the substation does.
According to construction	Classification based on where and how the substation is built.
According to voltage	Classification based on whether voltage is stepped up or stepped down.
Transformer, Switching, Converting	Examples of service-based substations.
Indoor, Outdoor, Pole-mounted	Examples of construction-based substations.
Step-up, Step-down, Distribution	Examples of voltage-based substations.

6 Major Equipment in a Substation

Equipment	Function
Power transformer	Steps up or steps down voltage.
Busbar	Common conductor to which incoming and outgoing circuits are connected.
Circuit breaker	Opens the circuit under normal and fault conditions.
Isolator	Provides visible isolation after circuit is made dead.
Current transformer CT	Steps down high current to a small measurable value for meters and relays.
Potential transformer PT	Steps down high voltage to a safe measurable value for meters and relays.
Lightning arrester	Protects equipment from lightning and switching surges.
Protective relay	Detects abnormal condition and sends trip signal to circuit breaker.
Earthing system	Provides safe path for fault current to ground.
Insulator	Supports live conductors and provides insulation from earth.
Battery and DC supply	Supplies control power for relay and circuit breaker tripping.
Capacitor bank	Improves power factor and voltage profile.

7 Single-Line Diagram of a Distribution Substation



Single-Line Diagram of 11 kV / 400 V Distribution Substation

How to Read the Diagram

Word/Symbol	Meaning
Incoming 11 kV Line	The high-voltage supply line entering the substation.
Lightning Arrester	Protects the substation from lightning and surge voltage.
Isolator	Provides visible isolation for maintenance. It should not be opened on load.
CT	Current transformer. It gives reduced current to meters and relays.
Circuit Breaker	Opens the circuit during fault or switching operation.
PT	Potential transformer. It gives reduced voltage to meters and relays.
Distribution Transformer	Steps down 11 kV to 400/230 V for consumers.
LT Busbar	Low-tension common conductor from which outgoing feeders are taken.
Feeder 1, 2, 3	Outgoing distribution lines supplying different load areas.
Single-Line Diagram	A simplified representation where three-phase system is shown by one line.

8 Busbar

Definition

A **busbar** is a common conductor used in a substation to collect electrical power from incoming circuits and distribute it to outgoing circuits.

A busbar is generally made of:

- copper, or
- aluminium.

Busbar is the common junction point of incoming and outgoing circuits.

9 Why Busbar Arrangement is Important

Busbar arrangement affects:

1. reliability,
2. flexibility,
3. maintenance,
4. cost,
5. fault isolation,
6. continuity of supply.

Memory Hook

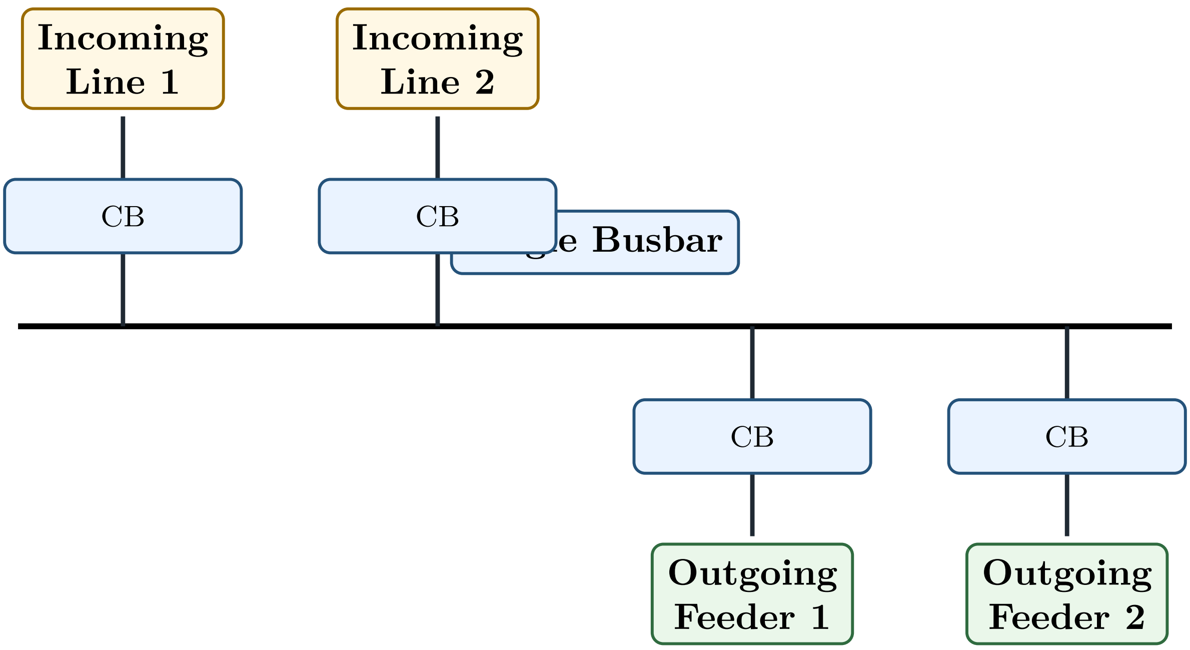
Simple busbar is cheap but less reliable. Complex busbar is costlier but gives better reliability and maintenance flexibility.

10 Single Busbar Arrangement

Definition

In a **single busbar arrangement**, all incoming and outgoing circuits are connected to one common busbar.

10.1 Diagram



Single Busbar Arrangement

How to Read the Diagram

Word/Symbol	Meaning
Single Busbar	One common conductor to which all circuits are connected.
Incoming Line 1, 2	Lines bringing power into the substation.
Outgoing Feeder 1, 2	Lines carrying power from substation to load areas.
CB	Circuit breaker used for switching and fault clearing of each circuit.
Single Busbar Arrangement	Simplest busbar scheme. It is cheap but bus fault causes complete shutdown.

10.2 Advantages

1. Simple construction.
2. Low cost.
3. Easy operation.
4. Less space required.

10.3 Disadvantages

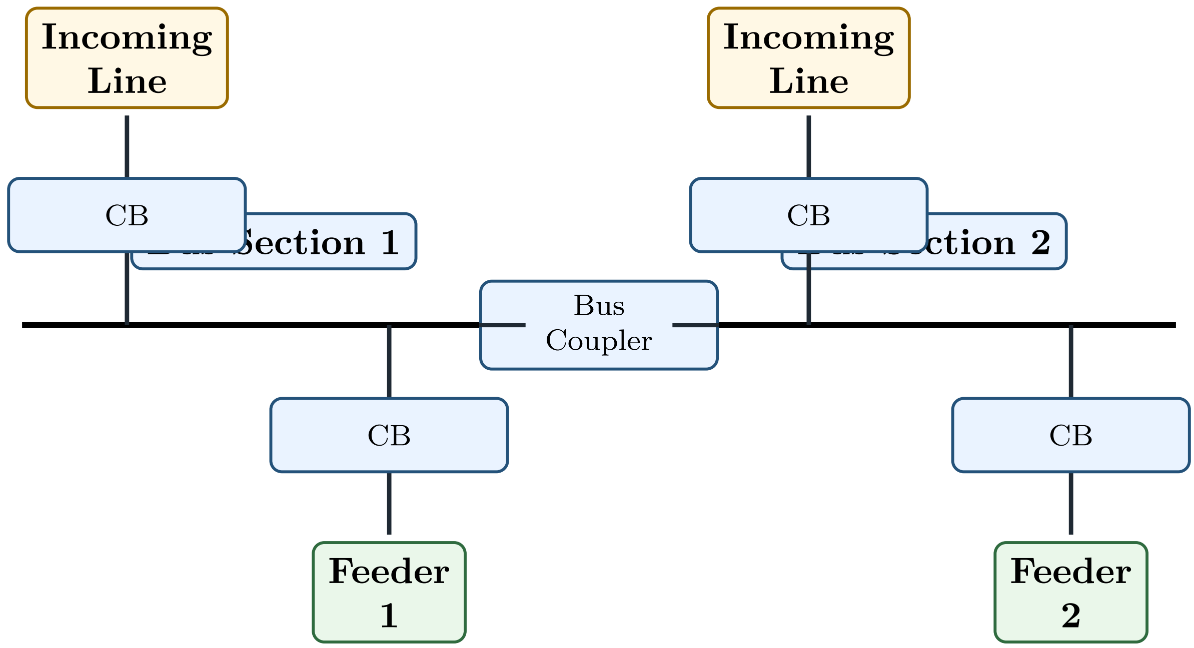
1. Fault on busbar shuts down the whole substation.
2. Maintenance of busbar requires complete shutdown.
3. Reliability is low.
4. Expansion is difficult.

11 Single Busbar with Sectionalisation

Definition

In a **single busbar sectionalised arrangement**, the busbar is divided into sections using a bus coupler circuit breaker.

11.1 Diagram



Single Busbar with Sectionalisation

How to Read the Diagram

Word/Symbol	Meaning
Bus Section 1 and 2	The main busbar is divided into two parts.
Bus Coupler	Circuit breaker connecting the two bus sections. It can join or separate them.
Incoming Line	Line bringing supply to a bus section.
Feeder 1 and 2	Outgoing lines supplying loads.
CB	Circuit breaker for switching and protection.
Sectionalisation	Dividing busbar into sections to improve reliability.

11.2 Advantages

1. Fault on one section does not necessarily shut down the whole substation.
2. Maintenance is easier than simple single busbar.
3. Reliability is improved.
4. Load can be transferred between sections using bus coupler.

11.3 Disadvantages

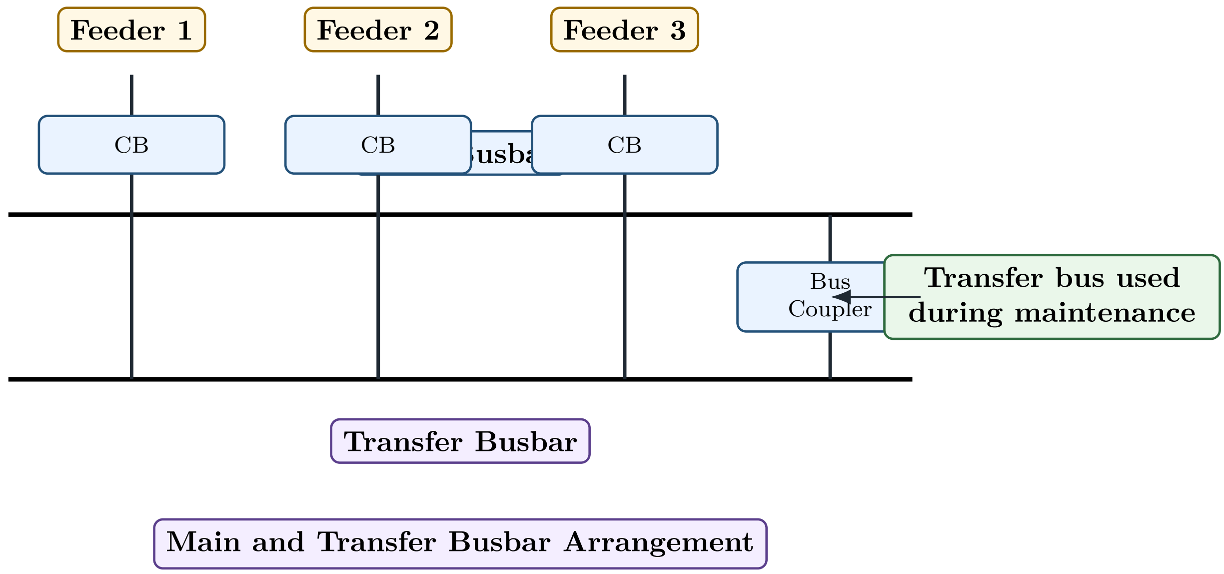
1. Cost is higher than simple single busbar.
2. Protection and operation are slightly more complex.

12 Main and Transfer Busbar Arrangement

Definition

In a **main and transfer busbar arrangement**, one main busbar normally carries load and an additional transfer busbar is used during maintenance of circuit breakers.

12.1 Diagram



How to Read the Diagram

Word/Symbol	Meaning
Main Busbar	Normal busbar that carries load in regular operation.
Transfer Busbar	Auxiliary busbar used when a circuit breaker requires maintenance.
Feeder 1, 2, 3	Outgoing or incoming circuits connected to bus system.
CB	Circuit breaker of each feeder.
Bus Coupler	Breaker used to connect main bus and transfer bus.
Transfer bus used during maintenance	It permits maintenance of a breaker without complete shutdown of the feeder.

12.2 Advantages

1. Circuit breaker maintenance is possible with less interruption.
2. Supply continuity is better than single busbar.
3. Flexible operation is possible.

12.3 Disadvantages

1. Cost is higher.

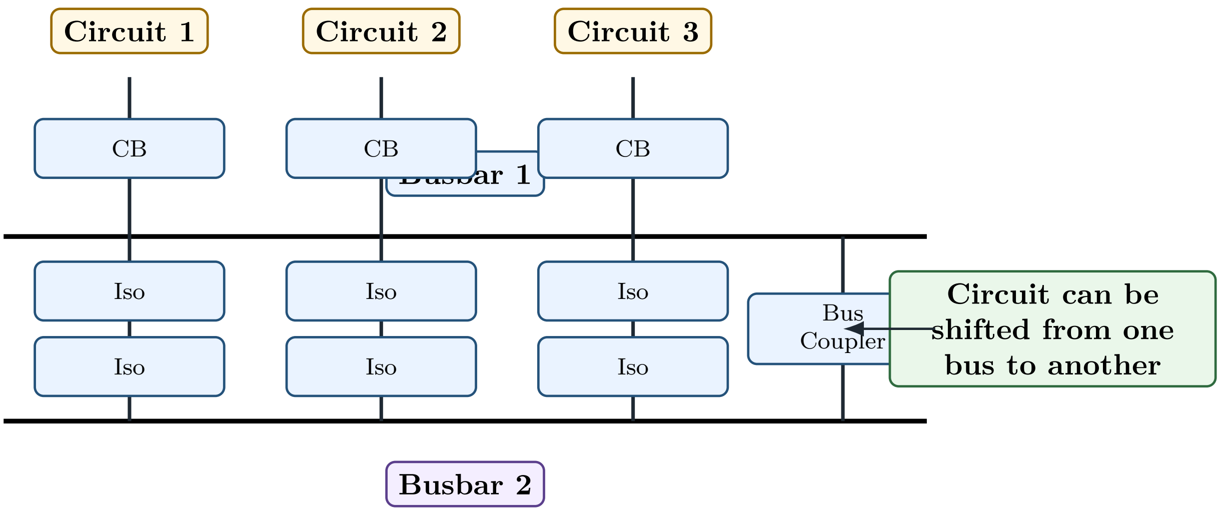
2. More isolators and switching operations are required.
3. Protection scheme becomes more complex.

13 Double Busbar Arrangement

Definition

In a **double busbar arrangement**, two busbars are provided and any circuit can be connected to either busbar.

13.1 Diagram



Double Busbar Arrangement

How to Read the Diagram

Word/Symbol	Meaning
Busbar 1	First main busbar.
Busbar 2	Second main busbar.
Circuit 1, 2, 3	Incoming or outgoing circuits.
CB	Circuit breaker for each circuit.
Iso	Isolator used to select connection to Busbar 1 or Busbar 2.
Bus Coupler	Circuit breaker used to connect or transfer load between two buses.
Circuit can be shifted	Any circuit can be transferred from one busbar to another for maintenance or operation.

13.2 Advantages

1. High flexibility.
2. Maintenance of one busbar is possible without total shutdown.
3. Circuits can be transferred from one bus to another.

4. Reliability is better than single busbar system.

13.3 Disadvantages

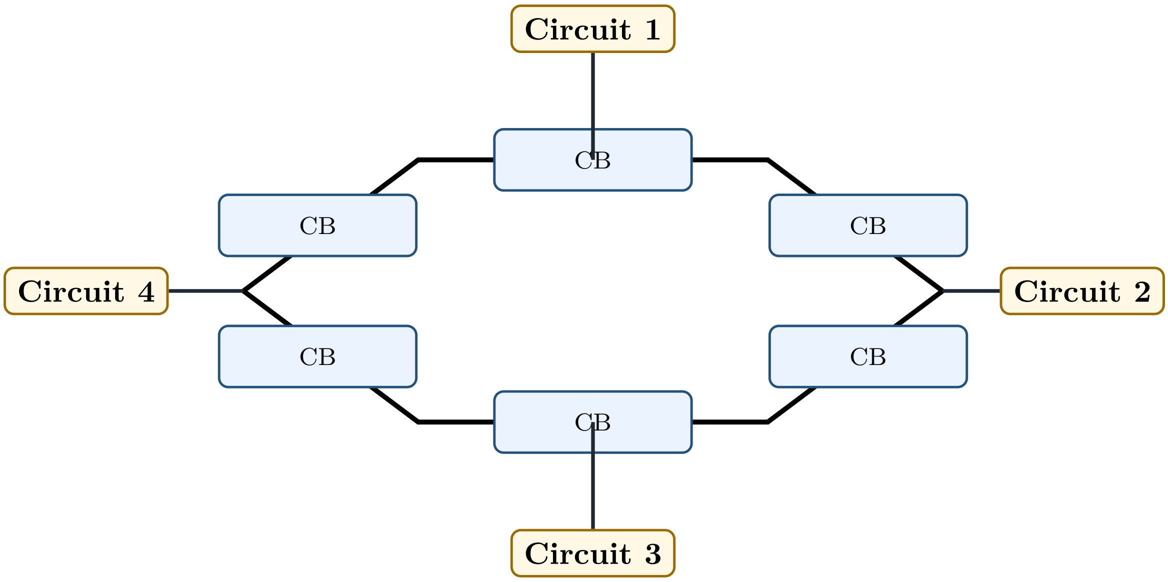
1. Cost is high.
2. More equipment is required.
3. Operation and protection are more complex.
4. More space is needed.

14 Ring Busbar Arrangement

Definition

In a **ring busbar arrangement**, circuit breakers are connected in the form of a ring and each circuit is tapped between two circuit breakers.

14.1 Diagram



Ring Busbar Arrangement

How to Read the Diagram

Word/Symbol	Meaning
Ring shape	Busbar and breakers are arranged in a closed loop.
CB	Circuit breakers placed around the ring.
Circuit 1, 2, 3, 4	Incoming or outgoing circuits connected to the ring busbar.
Ring Busbar Arrangement	Gives good reliability because a circuit can often remain supplied even if one breaker is out.

14.2 Advantages

1. High reliability.
2. Maintenance of breaker is possible with less interruption.
3. Fault on a section can be isolated.
4. No separate main busbar is required.

14.3 Disadvantages

1. Protection is complex.
2. Operation is more difficult.
3. Expansion of ring busbar is not very simple.

15 Pole-Mounted Substation

Definition

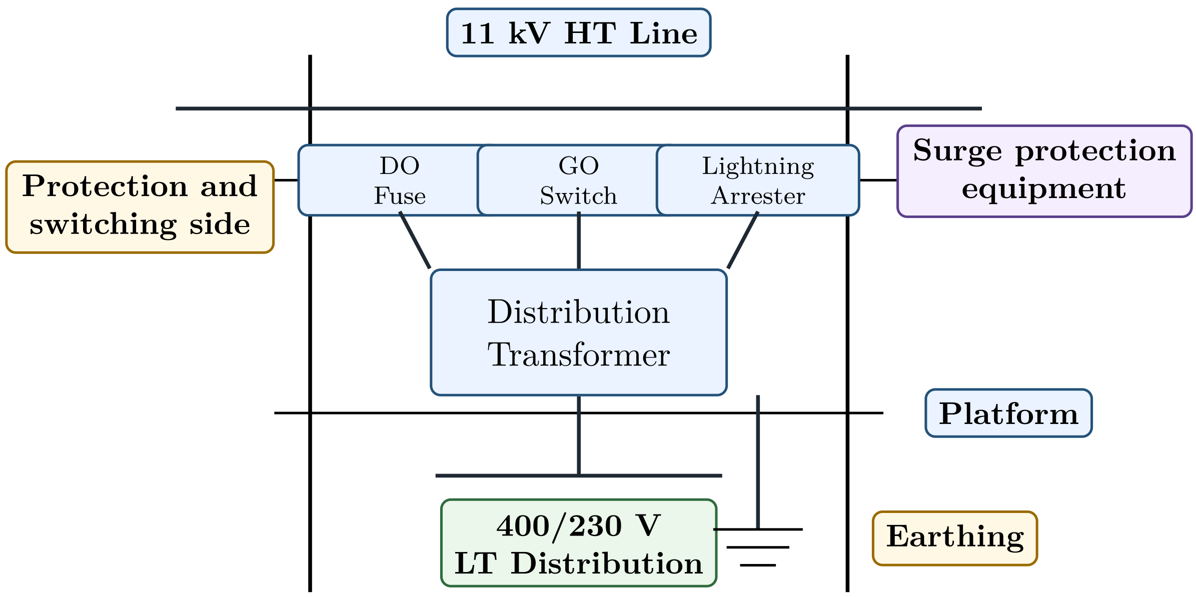
A **pole-mounted substation** is a small outdoor distribution substation in which the transformer and its accessories are mounted on poles.

It is commonly used for stepping down:

$$\boxed{11 \text{ kV} \rightarrow 400/230 \text{ V}}$$

for local distribution.

16 Diagram of Pole-Mounted Substation



Pole-Mounted Distribution Substation

How to Read the Diagram

Word/Symbol	**Meaning**
11 kV HT Line	High-tension line bringing 11 kV supply to the pole-mounted substation.
DO Fuse	Drop-out fuse used for protection of the transformer.
GO Switch	Gang-operated switch used for manual switching of the HT side.
Lightning Arrester	Protects transformer from lightning and surge voltage.
Distribution Transformer	Steps down 11 kV to 400/230 V.
Platform	Mechanical support for transformer mounted between poles.
400/230 V LT Distribution	Low-voltage output supplied to local consumers.
Earthing	Connection to ground for safety and protection.
Protection and switching side	Part containing fuse and switch arrangement.
Surge protection equipment	Lightning arrester diverts surge current to earth.
Pole-Mounted Distribution Substation	Outdoor pole structure used for local distribution.

17 Advantages and Disadvantages of Pole-Mounted Substation

17.1 Advantages

1. Low cost.
2. Simple construction.
3. Less space required.
4. Easy installation.
5. Suitable for rural and small load areas.
6. Natural air cooling is available.

17.2 Disadvantages

1. Suitable only for small capacity transformers.
2. Exposed to weather.
3. Maintenance may be difficult during bad weather.
4. Not suitable for large urban load centres.
5. Safety fencing or proper clearance is required.

18 CT and PT

18.1 Current Transformer

Definition

A **current transformer** or CT is an instrument transformer used to step down high current to a small safe value for measurement and protection.

Common secondary current values are:

$$\boxed{5\text{ A} \quad \text{or} \quad 1\text{ A}}$$

18.2 Potential Transformer

Definition

A **potential transformer** or PT is an instrument transformer used to step down high voltage to a small safe value for measurement and protection.

Common secondary voltage is:

$$\boxed{110\text{ V}}$$

18.3 Comparison Between CT and PT

Point	**CT**	**PT**
Full form	Current Transformer	Potential Transformer
Quantity transformed	Current	Voltage
Connection	Series with line	Parallel across line
Secondary value	Usually 5 A or 1 A	Usually 110 V
Use	Ammeter, relay current coil	Voltmeter, relay voltage coil
Important warning	Secondary should not be open-circuited while primary carries current	Secondary should not be short-circuited

Common Mistake

Important exam point:

$$\boxed{\text{CT secondary should never be open-circuited when primary current is flowing.}}$$

19 Lightning Arrester

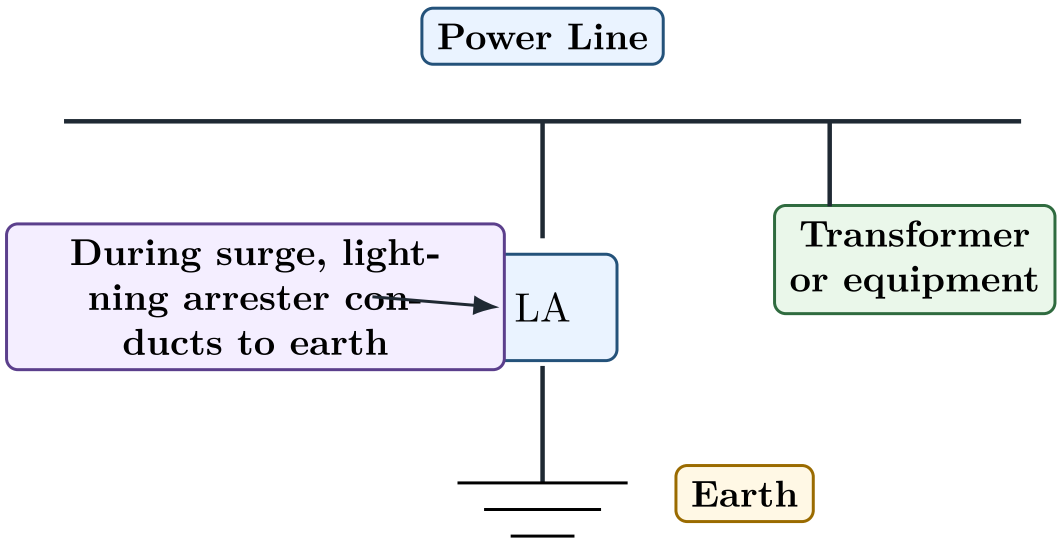
Definition

A **lightning arrester** is a protective device connected between line and earth to protect substation equipment from lightning and surge voltages.

During normal voltage, it does not conduct. During surge voltage, it conducts and diverts surge current to earth.

$$\boxed{\text{Lightning arrester diverts surge current to earth.}}$$

19.1 Lightning Arrester Diagram



How to Read the Diagram

Word/Symbol	Meaning
Power Line	Line on which lightning surge may appear.
LA	Lightning arrester. It provides low-resistance path to earth during surge.
Earth	Ground connection where surge current is safely discharged.
Transformer or equipment	Equipment protected by lightning arrester.
During surge	During abnormal high voltage, arrester conducts and protects equipment.

20 Comparison of Busbar Arrangements

Arrangement	Cost	Reliability	Main Use
Single busbar	Low	Low	Small substations
Single busbar sectionalised	Medium	Better	Medium substations
Main and transfer bus	Medium-high	Better	Breaker maintenance with less interruption
Double busbar	High	High	Important substations needing flexibility
Ring busbar	High	High	High reliability stations

Memory Hook

Busbar selection rule:

$$\boxed{\text{More reliability} \Rightarrow \text{more equipment} \Rightarrow \text{higher cost}}$$

21 Reverse Engineering of Semester Questions

Syllabus Word	Concept	Semester Question Form
Substation	Switching, transformation and protection centre	Define substation and state functions.
Classification of substation	Service, construction and voltage basis	Classify substations with examples.
Major equipment	Transformer, busbar, CT, PT, CB, isolator	List substation equipment and functions.
Single-line diagram	Simplified substation connection	Draw and explain distribution substation SLD.
Pole-mounted substation	Small outdoor distribution substation	Draw pole-mounted substation and explain parts.
Busbar arrangement	Common connection system	Explain single busbar or double busbar arrangement.
Bus sectionalisation	Bus divided into sections	Advantages of sectionalised busbar.
Main and transfer bus	Maintenance flexibility	Explain main and transfer busbar arrangement.
CT and PT	Instrument transformers	Compare CT and PT.
Lightning arrester	Surge protection	Explain lightning arrester with diagram.

22 Most Predictable Semester Questions from Phase 9

No.	Question	Marks
1	Define substation. State the main functions of a substation.	4–5
2	Classify substations according to service, construction and voltage.	5–6
3	List the major equipment used in a substation and state their functions.	6–8
4	Draw and explain the single-line diagram of a distribution substation.	6–8
5	What is a busbar? Explain its importance in substation.	3–4
6	Explain single busbar arrangement with advantages and disadvantages.	5
7	Explain single busbar sectionalised arrangement with diagram.	5–6
8	Explain main and transfer busbar arrangement.	5–6

9	Explain double busbar arrangement with advantages and disadvantages.	5–6
10	Draw and explain pole-mounted substation.	6–8
11	Compare CT and PT.	4–5
12	Explain lightning arrester with diagram.	4–5
13	Compare different busbar arrangements.	6

23 Exam Answer Templates

23.1 Template 1: Substation Theory Answer

Semester Answer Style

Answer sequence:

1. Define substation.
2. Write four main functions: transformation, switching, protection and distribution.
3. Classify substations if asked.
4. Mention major equipment.
5. Conclude with importance of reliability and control.

23.2 Template 2: Substation Equipment Answer

Semester Answer Style

Answer sequence:

1. Start with: A substation contains several electrical devices for transformation, switching and protection.
2. Write equipment names in a table.
3. Write one-line function of each equipment.
4. Give special attention to transformer, busbar, CB, isolator, CT, PT and lightning arrester.

23.3 Template 3: Busbar Arrangement Answer

Semester Answer Style

Answer sequence:

1. Define busbar.
2. Draw the required busbar scheme.
3. Label incoming lines, outgoing feeders, busbar, CB and bus coupler if present.
4. Explain normal operation.
5. Write advantages.
6. Write disadvantages.

23.4 Template 4: Pole-Mounted Substation Answer

Semester Answer Style

Answer sequence:

1. Define pole-mounted substation.
2. Write voltage level, usually 11 kV/400 V.
3. Draw two-pole structure.
4. Label HT line, DO fuse, GO switch, lightning arrester, transformer, LT output and earthing.
5. Write advantages and disadvantages.

24 Phase 9 Final Memory Sheet

One-Page Memory Sheet

1. Substation transforms, switches, protects and distributes power.
2. Transformer changes voltage level.
3. Busbar is a common conductor for incoming and outgoing circuits.
4. Circuit breaker interrupts fault current.
5. Isolator gives visible isolation but should not be opened on load.
6. CT steps down current for meters and relays.
7. PT steps down voltage for meters and relays.
8. Lightning arrester diverts surge current to earth.
9. Pole-mounted substation is used for small distribution transformers.
10. Single busbar is simple and cheap but less reliable.
11. Sectionalised busbar improves reliability.
12. Main and transfer busbar helps during breaker maintenance.
13. Double busbar gives high flexibility but high cost.
14. Ring busbar gives high reliability but complex protection.

25 Phase 9 Self-Test

Answer these without seeing notes:

1. What is a substation?
2. What are the main functions of a substation?
3. Classify substations according to service requirement.
4. Classify substations according to construction.
5. What is a busbar?
6. What is the function of circuit breaker in substation?
7. What is the function of isolator?

8. What is the function of CT?
9. What is the function of PT?
10. What is the function of lightning arrester?
11. Draw a single-line diagram of distribution substation.
12. Draw a pole-mounted substation.
13. Explain single busbar arrangement.
14. Explain sectionalised busbar arrangement.
15. Explain double busbar arrangement.
16. Compare CT and PT.

Common Mistake

Do not draw crowded substation diagrams. For exam diagrams, keep equipment in a straight sequence and write labels outside the conductor line.

End of Phase 9

Next Phase:
Phase 10 – Earthing, Wiring and Installation

Pipe earthing, plate earthing, factors affecting earth resistance, service mains, wiring systems, testing of installation and special lighting connections.